

## Brad's Autonomap

Software developer, Internet entrepreneur and self-driving car consultant Brad Templeton has launched the 'Autonomap' (<https://robocars.com/autonomap.html>), which lets users see the concentration of autonomous vehicles (AVs) on roads and sidewalks, in various parts of the world. It is his attempt to convince people that AVs have arrived.

This map shows all active autonomous vehicle services for both passengers and delivery. It has layers you can enable and disable. Markers in red represent services that are autonomous with no staff/safety-driver on board, and serving the public. Other colours represent different levels of autonomy. The small icons represent delivery vehicles, which cannot carry people onboard.

"This map is somewhat of a response to people who claim autonomous vehicles are something of the far future. As the aphorism goes, it's here, just not evenly distributed yet. However, it is largely limited to the USA and China, with a few areas in Europe, Canada, Korea and the Middle-East. This is due to the regulatory regimes in these areas. The USA has a permissive regulatory system—it's permitted until somebody forbids it. China isn't this way but can move quickly to change regulations," he explains in one of his articles about the map.



use delivery robots after graduating, if the service was available in their neighbourhood.

These friendly delivery bots are working their magic outside university campuses too. Some residential localities in the UK—such as Milton Keynes, Bedford, Northampton, Cambourne, Cambridge, and Leeds—have deployed autonomous deliveries.

According to a report, these robots are powered by renewable electricity, and can travel around three kilometres using lesser energy than it takes to boil water for a cup of tea. The report estimates that since their introduction in the UK, these autonomous robot deliveries have saved nearly 1.8 million kilometres of car journeys, with a 445-tonne reduction in carbon dioxide emissions. Starship's delivery robots have apparently completed around four million deliveries around the world to date.

Our fancy takes flight when we hear the phrase autonomous vehicle (AV). We start dreaming of cities as safe havens populated with self-driving cars and trucks. But a look at the roads around you would serve as a reality check, and you might lean more in favour of a report by Accenture, which indicates that there is no realistic chance that full-on self-driving will be available before 2030, and that too only in a tiny number of high-end sedans and SUVs! Given that roads and routes spell chaos and uncertainty, especially in developing nations, self-driving cars do seem far away from practical deployment.

But wait, all that research and tech is not going waste! Autonomous delivery systems, which pretty much use the same technologies as self-driving cars, have already arrived in various parts of the world. Just that,

they carry things instead of people. Autonomous delivery systems are being deployed in environments ranging from mines, war fields, and hospitals to warehouses, industries, airports, college campuses, and residential localities.

We have already touched upon their use in the industrial space in our story on cobots (collaborative robots). We have seen how they work alongside human workers, delivering spare parts, samples and many other goods, when and where required. In this story, let us explore the potential of autonomous last-mile delivery, which promises to be the near-future precursor to self-driving cars.

### Why the bot

The need for autonomous last-mile delivery was felt badly during the pandemic period when most merchants were open only for delivery but it